

Shamus Li

Research interests: computational imaging, generative priors for inverse problems, 3D/4D vision

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<https://shamus.li>

Education

PhD student in Computer Science, Cornell University 2024 – present

Advisor: Kristina Monakhova

MS in Electrical Engineering and Computer Sciences, Aug 2024
University of California, Berkeley

Advisor: Laura Waller

BS in Electrical Engineering and Computer Sciences, Aug 2023
University of California, Berkeley

Work Experience

Apple Sunnyvale, CA

Software Engineer Intern, Vision Products Group

May 2024 – Aug 2024

- Built Python tooling to automate QA testing for visionOS enterprise APIs, reducing end-to-end test execution time by more than 5x.
- Designed and implemented a task-driven camera simulation pipeline that generated terabytes of synthetic camera data across hundreds of scenes, enabling cross-functional teams to evaluate downstream vision algorithms without physical camera captures.
- Presented findings to Mike Rockwell and Vision Products Group leadership, informing the group's synthetic data strategy.

Lawrence Berkeley National Laboratory

Berkeley, CA

Research Intern

Oct 2022 – Aug 2023

- Built a real-time, human-in-the-loop pipeline for segmenting multiple structures in 3D soft X-ray cell tomograms, surpassing prior single-structure-only systems and reducing segmentation time from hours to minutes.
- Designed an adaptive brush in napari that let researchers annotate in 3D rather than slice by slice by conforming to organelle boundaries, improving precision and noise robustness over intensity-thresholding tools while reducing manual segmentation corrections.

Meta

Menlo Park, CA

VR Instructor

Jun 2023 – Aug 2023

- Developed and taught a 6-week VR experience design course for 30 students in partnership with Mission Bit, focusing on Unity, C#, 3D scene design, and interactive mechanics.

Palo Alto Research Center

Palo Alto, CA

Research Intern

Jun 2022 – Aug 2022

- Designed a workflow to efficiently visualize thermal properties in large buildings using multimodal imaging and sensor fusion. Built an augmented reality iOS app that improved sensor commissioning setup time by 80%.

Publications and Posters

When Do Diffusion Inverse Solvers Help in Lensless Imaging?

Shamus Li, John Thickstun, Kristina Monakhova

ICCP 2026, Spotlight Poster

Deep Optics Design for Extended-Field-of-View Depth-from-Defocus

Hasindu Kariyawasam, Shamus Li, Emma Alexander, Kristina Monakhova

ICCP 2026, Poster

Sparse Light Field Sampling Improves Casual 3D and 4D Reconstruction

Shamus Li, Ruiming Cao, Laura Waller, Kristina Monakhova, Sara Fridovich-Keil

ECCV 2026 S3DSGR Workshop, Spotlight Presentation

ICCP 2025, Spotlight Poster

QUTCC: Quantile Uncertainty Training and Conformal Calibration for Imaging Inverse Problems

Cassandra Tong Ye, Shamus Li, Tyler King, Kristina Monakhova

arXiv preprint (2507.14760)

CVPR 2025 UNCV Workshop, Spotlight Presentation

ICCP 2025 and 2026, Spotlight Posters

INSTANT: 3D Interactive Semantic Segmentation of Cellular Structure in Soft X-Ray Tomograms

Shamus Li, Saman Mostafavi, Axel Ekman, Valentina Loconte, Sookyoung Kim, Alejandro Brito, Carolyn Larabell

LBNL Molecular Biophysics & Integrated Bioimaging Workshop 2024, Poster

A Data-driven Method to Control Buildings Sub-zones to Spatio-temporal Profiles

Aayushman Sharma, Raman Goyal, Saman Mostafavi, Shamus Li, Alejandro Brito, Eric Bier

Building Simulation 2023

Sensorium: Commissioning Abundant Sensors with Augmented Reality and QR Codes

Eric Bier, Alejandro Brito, Raman Goyal, Saman Mostafavi, Les Nelson, Ritesh Sharma, Mahabir Bhandari, Niraj Kunwar, Shamus Li

Building Simulation 2023

Teaching Experience

Cornell University

- Teaching assistant for Kristina Monakhova in Computational Imaging (CS 6662), Fall 2024 and Foundations of Computational Imaging (CS 4660), Spring 2026
- Teaching assistant for Bharath Hariharan and Wei-Chiu Ma in Introduction to Computer Vision (CS 4670/5670), Spring 2025

University of California, Berkeley

- Teaching assistant for Nicholas Weaver in Great Ideas in Computer Architecture (CS 61C), Fall 2022
- Teaching assistant for John DeNero and Pamela Fox in Structure of Interpretation of Computer Programs (CS 61A), Fall 2021

Honors

Outstanding Teaching Assistant Award, Cornell CS Department, Spring 2026

ICCP Summer School Fellow (Received scholarship to attend the summer school and the conference), 2025

Officer, Eta Kappa Nu (HKN) Honor Society

Dean's List, UC Berkeley EECS Department

Service

Reviewer for ECCV 2026 S3DSGR Workshop; IEEE Transactions on Computational Imaging, 2025; Nature Photonics, 2024.

Cornell University Computer Science PhD admissions reader; 2025, 2026.

Cornell CS Student-Applicant Support Program mentor and Visit Days organizer; 2025, 2026.